Experiment Setup for 2026 Eclipse Shadow Bands

Version 1.0

Before You Participate

This project is a voluntary citizen science activity. By participating, you acknowledge that you are responsible for your own safety, travel, equipment, and choice of observation location. You agree to follow all applicable laws and to use proper eclipse eye protection whenever viewing the Sun.

Participation may involve outdoor activities, uneven terrain, changing weather conditions, and the use of electronic equipment. By taking part in this project, you voluntarily assume all risks associated with your participation and release the project organizers, volunteers, sponsors, collaborators, and affiliated organizations from liability for any injury, illness, property damage, loss, or other claims arising from or related to your participation, except where prohibited by law.

If you do not agree to these terms, do not participate.

Experience Something Rare

During the few moments before and after totality, rippling patterns called shadow bands race across the ground[1]. This citizen science project invites eclipse observers across Iceland and Spain to record shadow bands and help answer questions that remain largely unexplored. No prior scientific experience is required. If you can follow these instructions, you can contribute valuable data.



Figure 1: Eclipse shadow band experimental setup at the 2024 Eclipse[2]. This image contains shadow bands that, while difficult to see in the image, are detectable by the human eye during an eclipse and become clear after image processing.

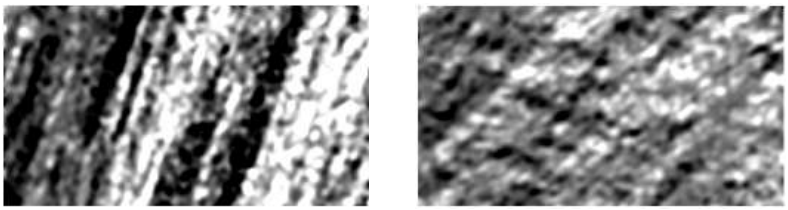


Figure 2: Eclipse shadow bands extracted from a video recording through an image processing framework[2]. The left image shows bands that are mostly aligned in the same orientation, but the bands on the right image are multimodal with two sets of bands are traveling in orthogonal directions simultaneously!

What You Need

- 1 smartphone with GPS, compass, and vertical-level applications (these are available on most smartphones by default).
- 1 smartphone (same as above) or camera with HD video recording capability at 60 frames per second with storage for at least 30-minutes of video (approximately 30 GB)
- 1 stand for smartphone or camera.

- 1 white cloth bed sheet or 4 white poster boards.
- 1 roll of clear packing tape (clear Scotch Heavy Duty shipping tape or equivalent).
- 5 small to medium cardboard boxes (approximately 20 inches or 40 cm in width).
- 1 ruler or tape measure to measure final board dimensions.



Select Your Observation Location

- See the following eclipse map.

Experiment Location 1: Edge of centerline

- **Purpose:** This experiment location will test if shadow band rotation correlates with centerline proximity[2].
- **Choose:** A location that is as far as possible from eclipse centerline but still within the totality band.
- **Examples:** Bilbao, Spain or Zamora, Spain or equivalent.

Experiment Location 2: High altitude

- **Purpose:** This experiment location will test if shadow band frequency and power diminish with altitude[2].
- **Choose:** A location that is somewhere high in altitude, like on top of a mountain.
- **Examples:** Pico San Millán, Spain or Moncayo, Spain or equivalent.

Experiment Location 3: Along eclipse centerline

- **Purpose:** This experiment location will provide a baseline for experiments 1 and 2 above.
- **Choose:** A location that is along eclipse centerline.
- **Examples:** La Pola, Spain or Salas, Spain or equivalent.

Arrive at Observation Location

- **Warning: The sun will be close to sunset during the eclipse, so you need to select a location with a clear view of the horizon.**
- Arrive at the location a few hours before totality to give yourself plenty of time to find a good location.
- Find a location that has a clear view of the west – northwest horizon.
- A good location might be on top of a hill away from trees or buildings.

Assemble Board at Observation Location

- Attach four cardboard boxes together with overlap and tape to form a large board.



- Stretch the white sheet over assembled board and secure with tape on the back.



- Assemble the fifth box and place a weight inside, like a bag.
- Prop the board against the fifth box so it faces sun during your setup about 2-hours before totality.
- For observers in Iceland, the sun will be about 25 degrees above the horizon, so you can position the board at an angle using the weighted box.
- For observers in Spain, the sun will be very close to sunset and near the horizon, so you will want to set up the board to be near vertical.
- The board does not need to be perfectly aligned with the sun. Keep the board straight and flat.



- Secure the overall structure firmly so it does not move in the wind.

- Maintain straight edges and right angles. This allows the image processing software to accurately correct camera perspective.
- Orient the board approximately perpendicular to the Sun so the shadow bands have maximum contrast.
- Place the camera stand in front of the board to the side so that the shadow does not fall on the board.



Record Experiment Setup Information

- Record the width and height of the board.
- Record your GPS latitude and longitude.
- Record angle of the board's front surface relative to the vertical direction (for Spain observers, ideally around 5 to 15 degrees depending on your location)
- Record angle of the board's bottom edge relative to the vertical direction (ideally around 90 degrees, flat on ground)



- Record compass orientation of the board's front surface. It should be in the Northwest direction.



Set Up the Camera

- Place the phone on a stable stand facing the board but slightly off to the side to avoid any shadows from the camera stand appearing on the board.
- Position the phone as close to the board as possible but far enough so that all four corners of the board are recorded.
- The post process image processing techniques can account for the camera being slightly off centered both laterally and vertically. It is more important to have a stable camera and board that will not move due to wind or other disturbances.
- Ensure the camera remains completely stationary throughout the recording.
- Use HD resolution and 60 frames per second.
- Turn off auto-focus if enabled.
- Do not change camera settings once recording has begun.
- Do not move camera once recording has begun.

Record Control Samples

- Record a 1-minute video sample prior to C1 (first contact). The exact time of C1 depends on your location. This should be about 90-minutes priority to totality. Note time of recording.
- Record a 1-minute sample after C1 (first contact) but before C2 (start of totality). This should be about 30 minutes prior to totality. Note time of recording.
- Use these control sample recordings to test your camera setup. If you make an error recording the control sample, start over.

Record Shadow Bands

- Begin continuous recording 10 minutes before totality (C2).
- Continue recording through totality and until at least 10 minutes after totality ends (C3).
- Do not touch the camera or the sheet during recording.
- Keep all equipment stable throughout the observation period.
- Shadow bands typically appear a few minutes before and after totality.
- Watch for faint, shimmering light and dark bands.
- They may resemble ripples, waves, or heat-shimmer patterns.
- The bands can move rapidly across the surface.

Avoid Common Mistakes

- Board bent and not flat

- Camera or board shaking in the wind.
- Autofocus changing during recording.
- Board partially shaded.
- Recording starts too late or ends too early.
- People walking in front of the board.

Share Your Data

- Share your video recordings and setup information with jconti@alum.mit.edu.
- Upload instructions will be provided after the eclipse.

Thank You for Contributing

Every successful observation helps improve our understanding of eclipse shadow bands. Even if your recording does not capture obvious shadow bands, your data remain scientifically valuable and should still be submitted.

References

1. https://en.wikipedia.org/wiki/Shadow_bands
2. https://arxiv.org/abs/2605.09621
3. https://nso.edu/for-public/eclipse-map-2026

Experiment Setup Information

	Measurement	Experiment Value
Location	Longitude	
	Latitude	
	Altitude	
Board Geometry	Width (cm)	
	Height (cm)	
Board Orientation	Angle of board facing surface relative to vertical direction	
	Angle of the board bottom edge relative to vertical direction	
	Orientation of board relative to compass true north	
Camera	Device type	
	Frames per second	
	Resolution	
Video recording	Control sample 1: start/end time	
	Control sample 2: start/end time	
	Eclipse: start/end time	